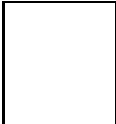




PROJECT AND TEAM INFORMATION

Project Title

Urban Emergency Pre-emption Engine (UEPE)

TeamName:	Avengers
Team member 1 (Team Lead)	Mohit Singh Basera 2027920 2010012056@geu.ac.in 
Team member 2	Pranav Pandey 2027984 2510011023@geu.ac.in 
Team member 3	Riddhi 2028472 2512520008@geu.ac.in 
	

PHASE-II PROJECT PROGRESS DESCRIPTION

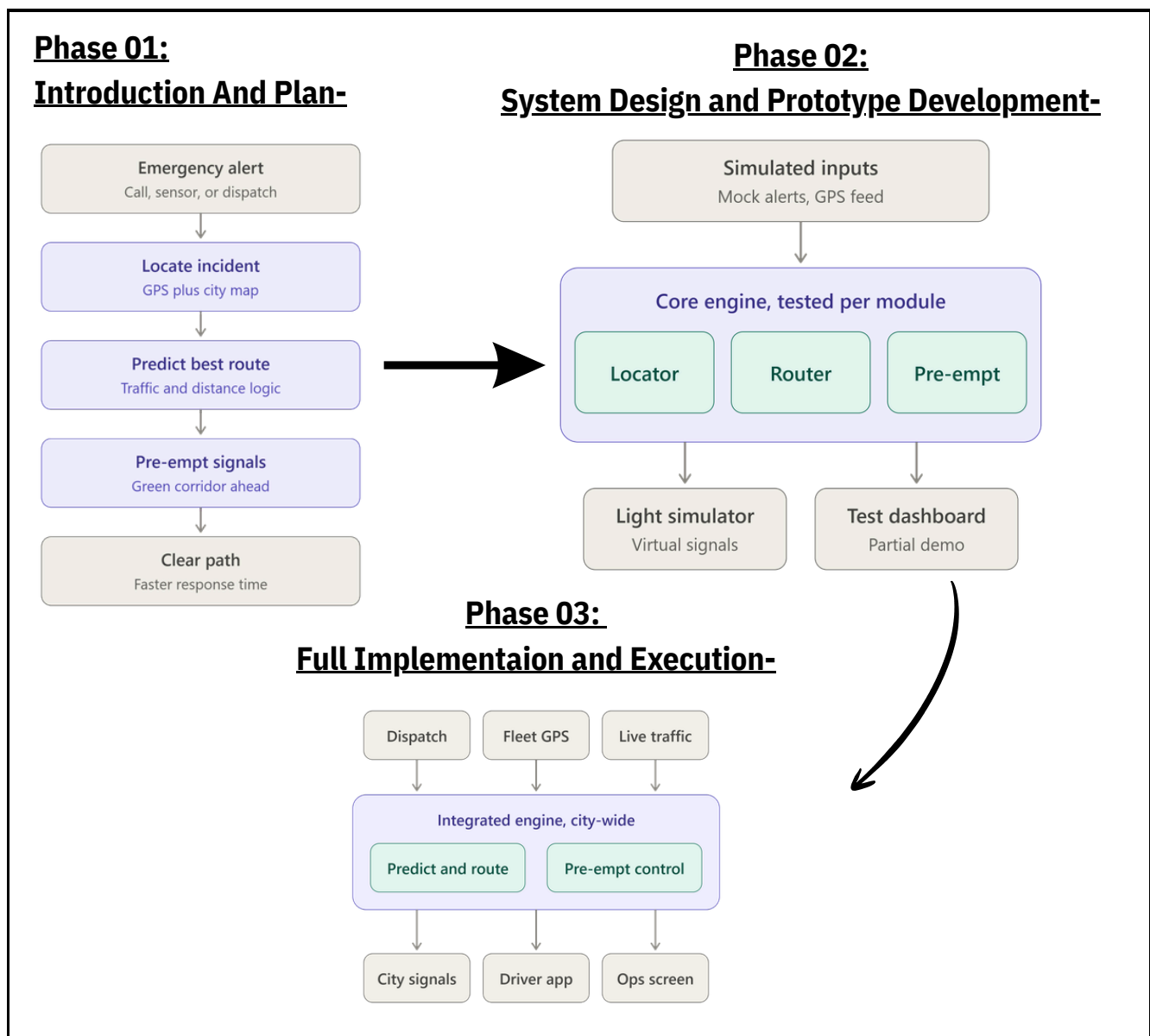
Project Abstract

The Urban Emergency Pre-emption Engine is a proposed smart traffic management to reduce the delay in emergency vehicles in urban areas. During emergencies traffic and red light might increase delay time. Our Project aims to address this problem by maintaining a priority to such vehicles at traffic interactions while maintaining normal traffic flow for other road users.

The system made by us will receive information about an emergency vehicle and decide it's direction, location and priority. Based on this information the engine will make a decision for upcoming traffic signal. Signal can temporarily change it's sequence to provide a safer and faster passage. Once the vehicle crosses the interaction, the system will restore the traffic signal to it's normal operation, reducing un-necessary disruption to regular traffic.

Also in this a test dashboard will display result and demonstrate a partial working prototype, So that early issues or glitches can be found and fixed without real city integration.

Updated Project Approach and Architecture



Tasks Completed

Task Completed	TeamMember
Problem Identified, Phase (01,02) Architecture Diagram, Concepts Identified, Working Understanding.	1.Mohit Sing Basera-2010012056 2.Pranav Pandey-2510011023 3.Riddhi-2512520008

Challenges/Roadblocks

- 1.Getting real time traffic, GPS and signal data
- 2.Integrating all modules with low latency.
- 3.Predicting the fastest route as the state of being very full of people or traffic changes.
- 4.Signal timing changes safely without conflicting green signals.
- 5.Not complete knowledge of the topics used in this project.
- 6.Keeping alert signal delay very low.
- 7.Handling multiple emergencies at the same junction.
- 8.Keeping system safe if a modules fails or a network fails (or is unavaliable).

Tasks Pending

Task Pending	Team Member (to complete the task)
Putting all modules together into 1 working system and testing everything properly, including failures and multiple emergencies, fixing the errors and get response time down.	1.Mohit Sing Basera-2010012056 2.Pranav Pandey-2510011023 3.Riddhi-2512520008

Project Outcome/Deliverables

- 1.The project delivers a working engine that clears the road for emergency vehicles. When an alert comes in, the system locates the incident, predicts the fastest route, and turns the traffic signals along it green ahead of the vehicle. Once the vehicle has passed, the signals return to normal.
- 2.The system is built from three tested modules: Locator, Router and Pre-empt controller. Each was first validated in a simulated environment with mock alerts, GPS feeds and virtual signals. They were then integrated into a full system with a driver app and an operations dashboard.
- 3.The main result is a measurable drop in emergency response time, shown by comparing trips with and without pre-emption. Safety is built in through a fail-safe mode that prevents conflicting greens and a priority rule for overlapping emergencies.
- 4.Every minute saved can change the outcome of a medical emergency or a fire. By automating what traffic police do manually, the engine makes responses faster and more consistent, and it lowers the risk of accidents at junctions.

Progress Overview

- Problem and objectives defined.
- Architecture diagrams made for all three phases.
- Project approach decided: modular build with a simulation before any real city data.

Still to do

- Finish the working prototype and test dashboard.
- Integrate all modules, test edge cases such as overlapping emergencies, and prepare the final report and presentation.
- Testing each module separately.
- Writing the test report with the problem list.
- Code for different modules is made.

Codebase Information

For the detailed information for the project you may consider the link below:

<https://github.com/Riddhi5101/PBL.git>

The **Urban Emergency Pre-emption Engine (UEPE)** is a smart traffic management system designed to reduce delays faced by emergency vehicles such as **ambulances, fire trucks, and police vehicles** at traffic intersections.

The system temporarily gives priority to the emergency vehicle and restores the traffic signal to its normal operation after the vehicle passes

Testing and Validation Status

TestType	Status (Pass/Fail)	Notes
Tests to be done yet	Tests to be done yet	Tests to be done yet

Deliverables Progress

- 1.Phase 2 architecture diagram:** a picture of phase 2 system.
- 2.Route algorithm:** a working shortest-path method (like Dijkstra or A*) that picks the best route.
- 3.Working modules:** a basic version of each of the three parts, tested one by one.
- 4.Presentation slides:** slides for the Phase 2 review.
- 5.Class or module diagram:** a chart of how the code is organised.